

Chapter 6 Assignment — Trigonometric Functions: Right Triangle Approach

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers in radicals and π when possible. Use degrees or radians as requested and state domain restrictions when relevant.

1. Topic 1: Angle Measure, Radians, and Arc Length

Strategy Notes. Convert between degrees and radians using $180^\circ = \pi$ radians. Central-angle arc length: $s = r\theta$ and sector area: $A = \frac{1}{2}r^2\theta$ with θ in radians. Find coterminal angles by adding/subtracting 2π (or 360°).

Worked Example. Convert 225° to radians and find the arc length on a circle of radius 9 units subtended by 225° .

Solution. $225^\circ = 225 \frac{\pi}{180} = \frac{5\pi}{4}$. With $r = 9$ and $\theta = \frac{5\pi}{4}$, the arc length is $s = r\theta = 9 \cdot \frac{5\pi}{4} = \frac{45\pi}{4}$ units.

Practice.

(a) Convert to the requested units and find a coterminal angle in the indicated range:

1. $\frac{7\pi}{6}$ radians $\rightarrow$ degrees; give a coterminal angle in $[0, 360^\circ)$.
2. $-135^\circ \rightarrow$ radians; give a coterminal angle in $[0, 2\pi)$.

(b) A wheel of radius 12 cm turns through 40° . Find the arc length and the area of the swept sector. Give exact answers.

Challenge.

- (a) A radar antenna makes 18 revolutions per minute. What is the angular speed in radians per second? If the antenna tip is 2.5 m from the center, what is the linear speed (m/s) of the tip?

2. Topic 2: Trigonometry of Right Triangles

Strategy Notes. In a right triangle with acute angle θ , use $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$. Use the Pythagorean Theorem to find the third side. Angles of elevation/depression are measured from horizontal.

Worked Example. A 15-ft ladder leans against a wall making a 68° angle with the ground. How high up the wall does it reach, and how far is the base from the wall?

Solution. Height $= 15 \sin 68^\circ \approx 13.9$ ft. Base distance $= 15 \cos 68^\circ \approx 5.6$ ft.

Practice.

- (a) In $\triangle PQR$ with $\angle PQR = 90^\circ$, $\angle QRP = 41^\circ$, and $QP = 10$. Find PR and QR to the nearest hundredth.
- (b) From a point on level ground 160 m from a building, the angle of elevation to the roof is 28° . Estimate the building height to the nearest meter.

Challenge.

- (a) A kite string is 120 ft long and makes a 34° angle with the ground. If the kite is 6 ft above the user's hand, estimate the horizontal distance from the person to the point on the ground below the kite (nearest foot). Assume the string is taut and straight.

3. Topic 3: Trigonometric Functions of Angles

Strategy Notes. For an acute angle θ in a right triangle, evaluate the six trigonometric functions using side ratios. For a non-vertical line $y = mx + b$, the acute angle α that the line makes with the positive x -axis satisfies $\tan \alpha = |m|$ and $\alpha = \arctan(|m|)$.

Worked Example. If $XY = 3$ and $YZ = 4$ in right triangle $\triangle XYZ$ with right angle at Y , find $\sin \angle X$, $\cos \angle X$, and $\tan \angle X$.

Solution. The hypotenuse $XZ = 5$ by 3-4-5. Thus $\sin \angle X = \frac{\text{opp}}{\text{hyp}} = \frac{4}{5}$, $\cos \angle X = \frac{3}{5}$,
 $\tan \angle X = \frac{4}{3}$.

Practice.

- (a) Given a right triangle with legs in the ratio $1 : \sqrt{3}$ and hypotenuse 8, find the acute angles and all six trigonometric functions of the smaller angle.

- (b) Find the measure (nearest degree) of the acute angle formed by the x -axis and the line $3x - 4y = 10$.

Challenge.

- (a) In right triangle ABC with right angle at C , suppose $\sin A = \frac{5}{13}$. Find $\cos B$ and $\tan B$.

4. Topic 4: Inverse Trigonometric Functions and Right Triangles

Strategy Notes. Principal values: $\arcsin x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arccos x \in [0, \pi]$, $\arctan x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Use right-triangle diagrams to translate between inverse trig expressions and algebraic forms.

Worked Example. Evaluate $\arcsin(-\frac{\sqrt{2}}{2})$ and $\arctan(-1)$.

Solution. $\arcsin(-\frac{\sqrt{2}}{2}) = -\frac{\pi}{4}$ and $\arctan(-1) = -\frac{\pi}{4}$, both lying in their principal ranges.

Practice.

(a) Compute exactly: $\arccos(-\frac{1}{2})$, $\arcsin(\frac{\sqrt{3}}{2})$, $\arctan(\sqrt{3})$.

(b) Let $\theta = \arcsin\left(\frac{2x}{\sqrt{1+4x^2}}\right)$ with $x \geq 0$. Find $\cos \theta$ in terms of x .

Challenge.

- (a) A ramp rises 30 inches over a horizontal run of 8 feet. Find the angle of inclination in degrees to one decimal place and express it using an inverse trigonometric function.

5. Topic 5: The Law of Sines

Strategy Notes. In any triangle with sides a, b, c opposite angles A, B, C , respectively, $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$. Use it for AAS/ASA (and SSA with care). For SSA, check for 0, 1, or 2 solutions by comparing the altitude $h = b \sin A$ with side a .

Worked Example. In $\triangle ABC$, $A = 42^\circ$, $a = 8$, and $b = 10$. Find B and c (nearest tenth). Determine whether the solution is unique.

Solution. $\sin B = \frac{b \sin A}{a} = \frac{10 \sin 42^\circ}{8} \approx 0.837$. Then $B \approx 56.7^\circ$ (valid) and $B' \approx 123.3^\circ$ (rejected since $A + B' > 180^\circ$). Thus $B \approx 56.7^\circ$ and $C \approx 81.3^\circ$. Finally $c = \frac{\sin C}{\sin A} a \approx \frac{\sin 81.3^\circ}{\sin 42^\circ} \cdot 8 \approx 11.9$.

Practice.

(a) Solve $\triangle ABC$ given $A = 31^\circ$, $B = 72^\circ$, and $b = 17$. Find c and a .

(b) SSA case: $A = 44^\circ$, $a = 12$, $b = 15$. Determine the number of possible triangles and solve all that exist (nearest tenth).

Challenge.

- (a) From the top of a lighthouse, the angles of depression to two buoys located along the same line from the base are 6° and 12° . If the lighthouse is 40 m tall, find the distance between the buoys (nearest meter). Assume level sea.

6. Topic 6: The Law of Cosines and Heron's Formula

Strategy Notes. Law of Cosines: $c^2 = a^2 + b^2 - 2ab \cos C$ (and cyclic variants) for SSS or SAS. Use it to decide whether an angle is acute/right/obtuse. Area from sides a, b, c can be found by Heron's formula $[\triangle] = \sqrt{s(s-a)(s-b)(s-c)}$ with $s = \frac{1}{2}(a+b+c)$.

Worked Example. In $\triangle ABC$, $a = 7$, $b = 9$, and $C = 62^\circ$. Find c and the area of the triangle.

Solution. $c = \sqrt{7^2 + 9^2 - 2 \cdot 7 \cdot 9 \cos 62^\circ} \approx 8.4$. Area $= \frac{1}{2}ab \sin C = \frac{1}{2} \cdot 7 \cdot 9 \sin 62^\circ \approx 27.8$ square units.

Practice.

- (a) Two hikers leave the same point. One walks 4.2 mi on a bearing N35°E, the other 3.5 mi on a bearing S50°E. How far apart are they to the nearest tenth of a mile?
- (b) Given an SSS triangle with sides 8, 11, and 14, determine the largest angle and the area (nearest tenth). Use Heron's formula for the area.

Challenge.

- (a) Two planes depart simultaneously from an airport. One flies due west at 240 mph; the other flies 30° north of east at 320 mph. How far apart are they after 1.5 hours (nearest mile)?